

LAUREN SABO

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Education

MS Bioinformatics, Georgia Institute of Technology
BS Computer Science, Georgia Institute of Technology

Dec. 2026
Dec. 2025

Skills

Computing Languages & Software

Python (NumPy, pandas, PyTorch), Java, React/ Node.js,
Bash, Conda, Git, Kubernetes, AWS, SQL, HTML/ CSS,
R, C++, C, Google Cloud Platform (GCP), Docker

Bioinformatics

NCBI BLAST, Genome Assembly Software (HiFi.asm,
Flye, LJA), Genome Analysis Tool Kit (GATK),
BRAKER3, NGS, SAMtools

Data Visualization

D3, Tableau, matplotlib, plotly

AI Tools

Claude Opus 4.6, Gemini 3.6 Flash, GPT-5.5

Work Experience

Graduate Research Assistant, Parker H. Petit Institute

Aug. 2025 — Present

- Engineering a robust QTL mapping pipeline for ultra-low coverage (~0.1X) sequencing data, enabling reliable detection of sex-linked loci in *Aulonocara sp.* “Yellow Head.”
- Applying probabilistic genotype inference with **Hidden Markov Models** to accurately impute offspring haplotypes, pinpointing candidate regions on LG10 that underlie male determination

AI Engineer Intern, Cornerstone OnDemand

May 2026 — Aug. 2026

- Integrated backend systems into an MCP-based Compensation Agent for Paramount demo, to power flight-risk identification and data-driven compensation recommendations for managers and HR
- Deployed and maintained agent infrastructure on **AWS** using **Kubernetes**
- Designed and built a reusable template UI for the Compensation Agent, establishing a scalable foundation for future client deployments across **Cornerstone AI**, the company's new AI product suite

Co-founder & Chief of Product Design, Quuri Co.

May 2024 — Aug. 2026

- Led end-to-end product design and development of a research networking platform, translating researcher needs into an intuitive interface that improved cross-disciplinary collaboration.
- Engineered cloud-based full-stack systems using Firebase and **Google Cloud Platform (GCP)** to support real-time data updates and growth across academic research communities.

Undergraduate Research Assistant, Parker H. Petit Institute

May 2022 — Aug. 2025

- Accelerated genome assembly and scaffolding workflows for African cichlid species by developing optimized Python pipelines tailored for long-read sequencing data (**PacBio HiFi**).
- Improved genomic data accuracy and assembly depth through rigorous quality control, read filtering, and alignment, strengthening downstream comparative and evolutionary analyses.

Projects

FinPhaser: A Reproducible Pipeline for Local Ancestry and IBD Inference

Jan. 2026 — May 2026

- Generalized a local-ancestry inference and identity-by-descent (IBD) detection workflow from the McGrath Lab's *Aulonocara* × *M. conophoros* backcross project into a standalone, reproducible **Nextflow** pipeline
- Orchestrated a multi-tool workflow (HMM-based ancestry inference via Ancestry_HMM, IBD detection via SPORE/TRUFFLE) operating on **VCF-formatted genotype calls**, with per-process **Conda** isolation, reducing execution to a single command

“Comprehensive Structural and Functional Genome Annotation of Lake Malawi Cichlids

Jan. 2025 — May 2025

Using BRAKER3: Insights into Evolutionary Adaptation and Speciation.”

- In accordance with the President’s Undergraduate Research Award (2025)
- Advanced comparative genomics of five African cichlid species by integrating PacBio HiFi assemblies and **RNA-Seq data** in a **BRAKER3**-based annotation framework to detect lineage-specific adaptive genes.

Chattahoochee River Conditions App

Feb. 2024

- Built a real-time web application integrating **USGS Water Quality API data** to visualize river conditions, improving safety awareness and operational planning for the Georgia Tech Rowing Club.